

BEE 208 — Question 9: Count Passed Component Tests

Control Statement: for loop with if...else | Faculty of Engineering, ATU

1. ALGORITHM

Step 1: Start

Step 2: Declare variables result, passCount, failCount, i (int)

Step 3: Set passCount = 0, failCount = 0

Step 4: Use a for loop from i = 1 to 15

Step 5: Prompt user to enter 1 (pass) or 0 (fail) for component i

Step 6: Read result

Step 7: If result == 1, increment passCount by 1

Step 8: Else increment failCount by 1

Step 9: End loop

Step 10: Display passCount and failCount

Step 11: End

2. PSEUDOCODE

```
BEGIN

DECLARE result, passCount, failCount, i AS INTEGER

SET passCount = 0, failCount = 0

FOR i = 1 TO 15 DO

    OUTPUT 'Component', i, 'Result (1=Pass, 0=Fail):'

    INPUT result

    IF result == 1 THEN

        passCount = passCount + 1

    ELSE

        failCount = failCount + 1

    END IF

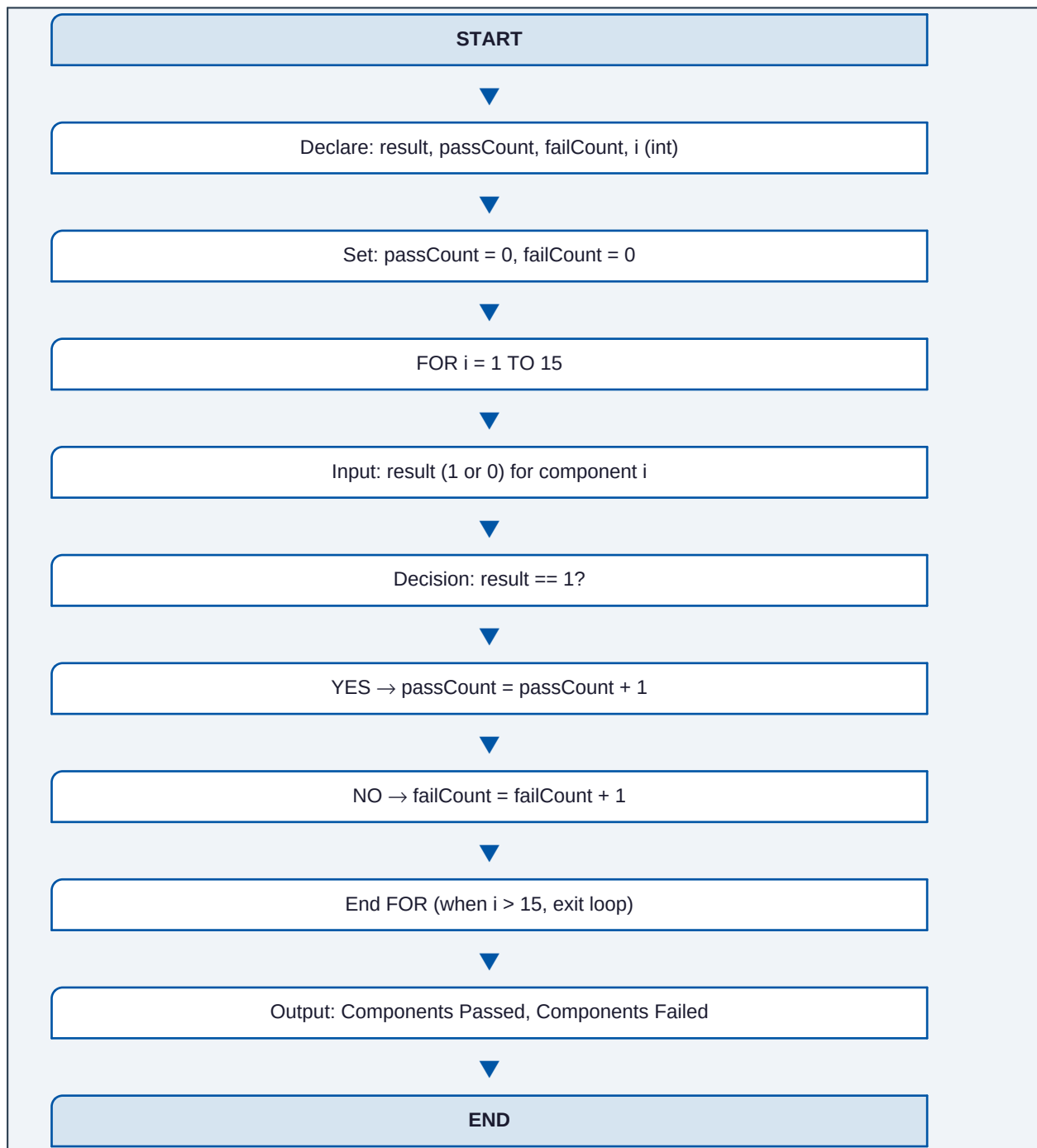
END FOR

OUTPUT 'Components Passed:', passCount

OUTPUT 'Components Failed:', failCount

END
```

3. FLOWCHART (Structured Diagram)



4. TEST RESULTS

Test Item	Value / Description	Test Item	Value / Description
Input	15 values: 10 ones and 5 zeros	Expected Output	Passed=10, Failed=5
Input	15 values: all ones	Expected Output	Passed=15, Failed=0